

CONTROL STRUCTURES

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CONTROL STRUCTURES

Sequential Logic

Selection Logic

- *One Way Selection*
- *Two Way Selection*
- *Multiple Selection*

Iteration Logic - *Repetition*

C# CONTROL STRUCTURES

- A way to specify flow of control in any algorithm or program can be more clear and understood
- It basically analyzes and chooses in which direction a program flows based on certain parameters or conditions

CONTROL STRUCTURES

Sequential Logic

- It follows a serial or a sequence flow
- *Flow depends on the series of instruction*
- *Modules are executed in the obvious sequence*

CONTROL STRUCTURES

Sequential Logic



BASIC PROGRAM FLOW

- `x = 10` `# Step 1: assign 10 to x`
- `y = 20` `# Step 2: assign 20 to y`
- `z = x + y` `# Step 3: add x and y, and store the result in z`
- `print(z)` `# Step 4: print the result (30)`

CONTROL STRUCTURES

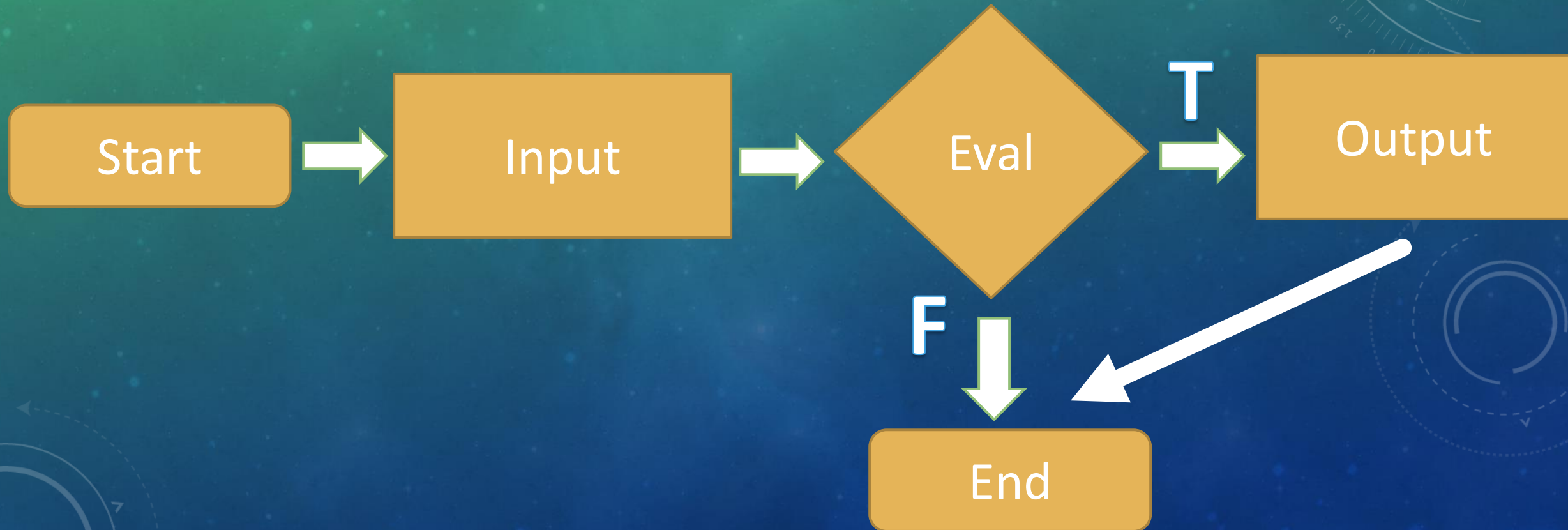
Selection Logic

- *Allows one set of statements to be executed if a condition is true and another set of actions to be executed if a condition is false.*

CONTROL STRUCTURES

Selection Logic – *One Way Selection*

Selection Logic – *One Way Selection*

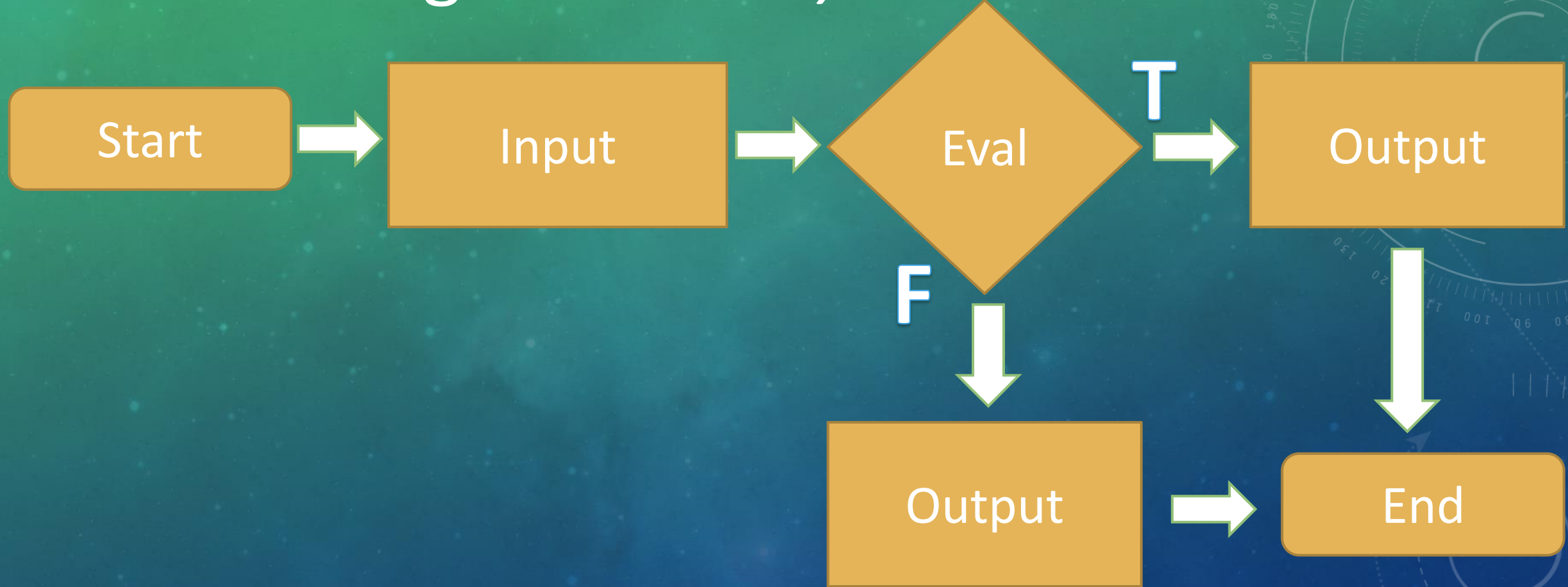


```
temperature = 30  
if temperature > 25:  
    print("It's a hot day!")
```

CONTROL STRUCTURES

Selection Logic – *Two Way Selection*

Selection Logic – *Two Way Selection*

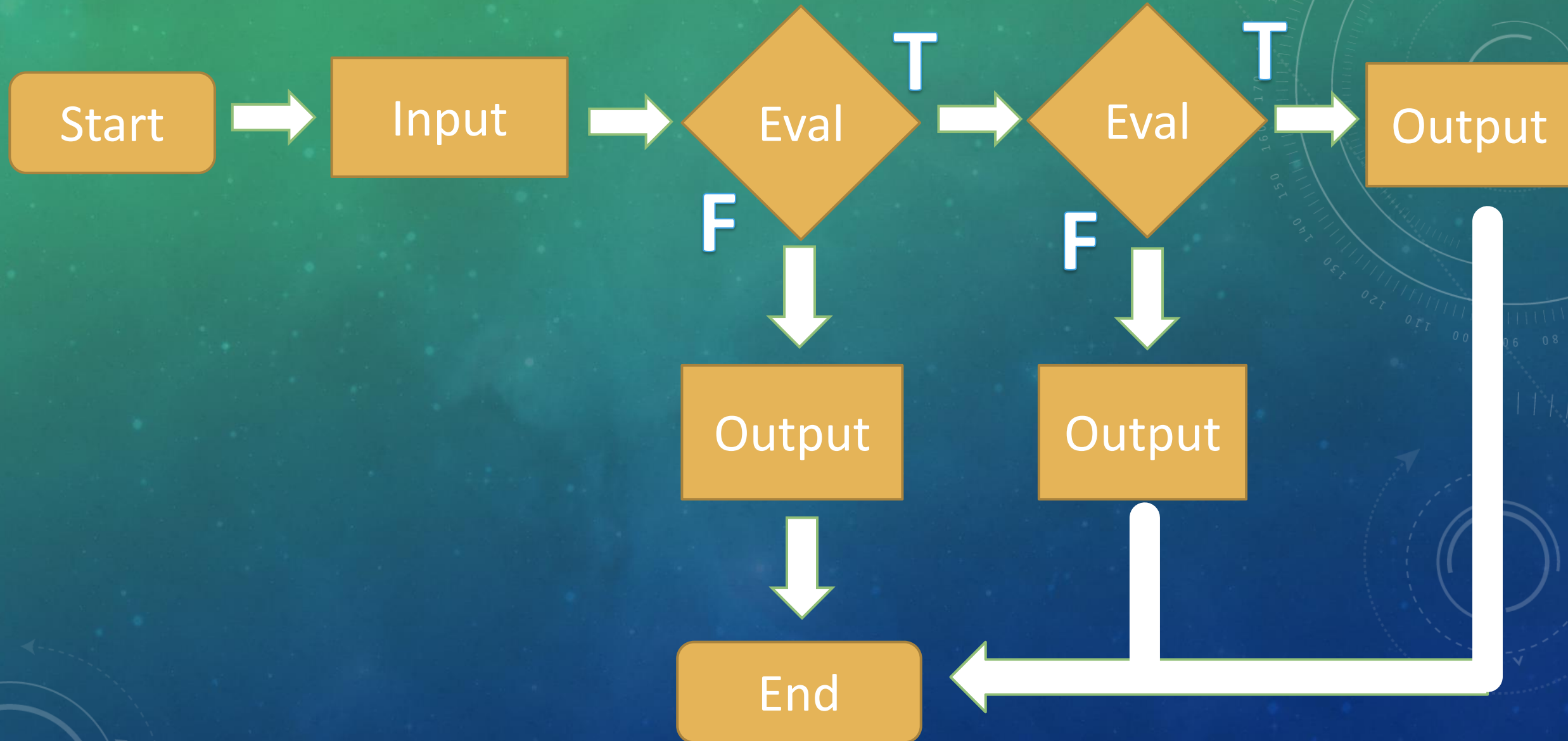


```
temperature = 20
if temperature > 25:
    print("It's a hot day!")
else:
    print("It's not a hot day.")
```

CONTROL STRUCTURES

Selection Logic – *Multiple Selection*

Selection Logic – *Multiple Selection*



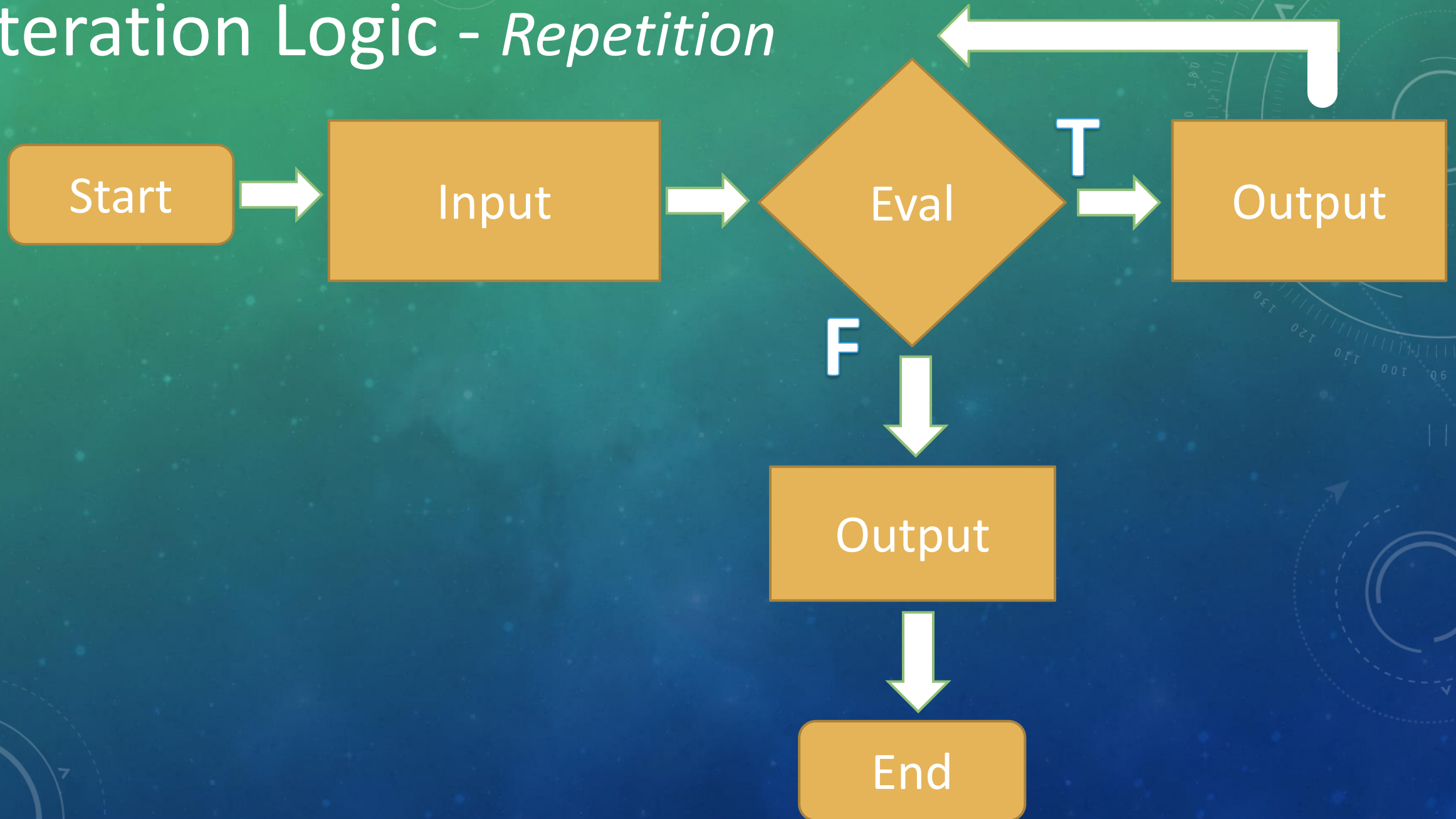
```
temperature = 15
if temperature > 30:
    print("It's a very hot day!")
elif temperature > 20:
    print("It's a warm day.")
else:
    print("It's a cool day.")
```

CONTROL STRUCTURES

Iteration Logic - *Repetition*

- Employs a loop which involves a “REPEAT Statement”

Iteration Logic - *Repetition*



FOR LOOP:

A FOR LOOP REPEATS A BLOCK OF CODE A SPECIFIC NUMBER OF TIMES OR ITERATES OVER A COLLECTION OF ITEMS (LIKE LISTS OR ARRAYS).

```
for i in range(5):  
    print(i)
```

0

1

2

3

4

WHILE LOOP:

A WHILE LOOP REPEATS A BLOCK OF CODE AS LONG AS A GIVEN CONDITION IS TRUE. IT IS TYPICALLY USED WHEN THE NUMBER OF ITERATIONS IS NOT KNOWN IN ADVANCE.

```
i = 0  
while i < 5:  
    print(i)  
    i += 1
```

0
1
2
3
4

CONTROL STRUCTURES

Sequential

- *Do*

One-Way Selection

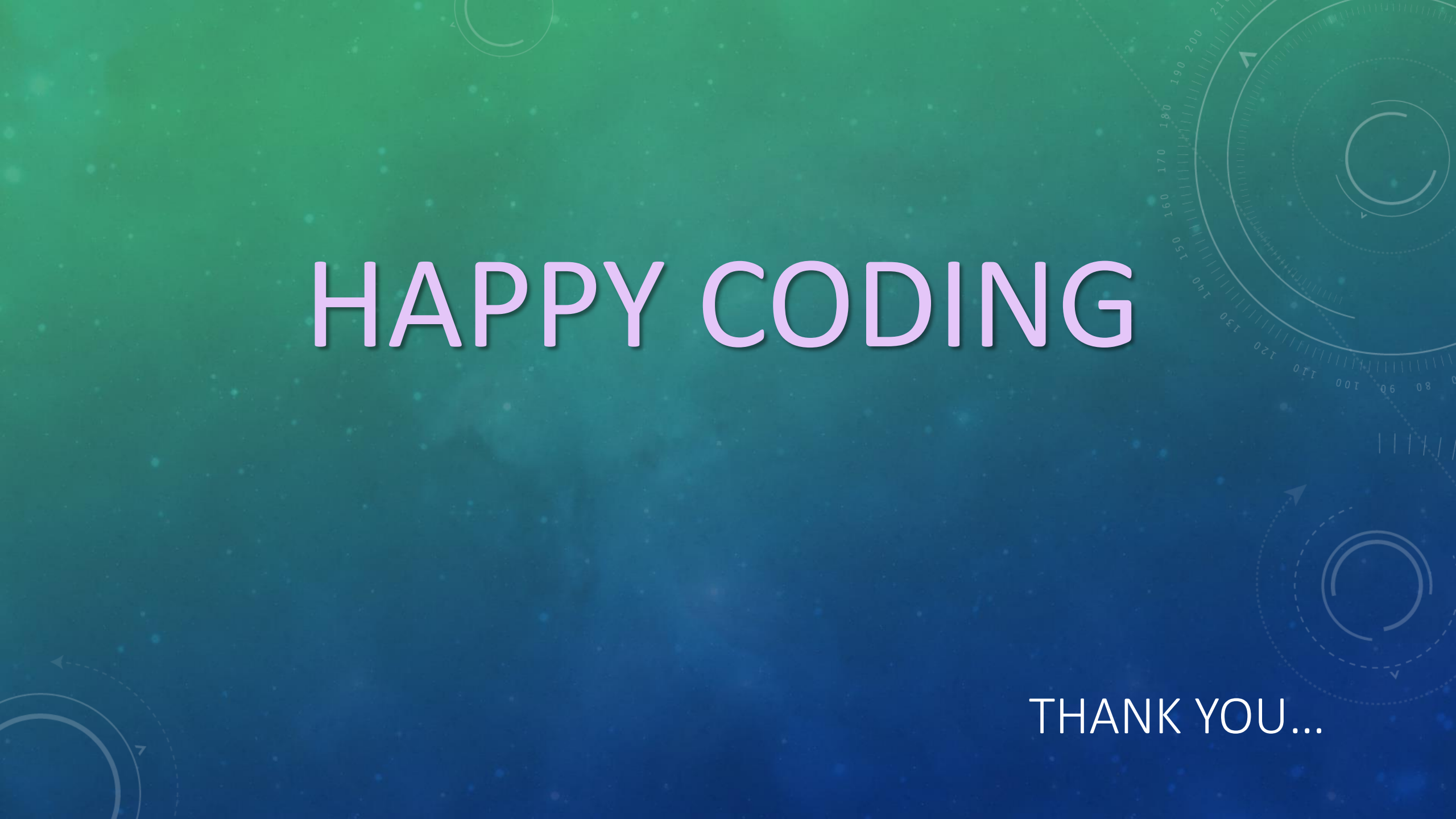
- *Decide*

Two Way Selection

Multiple Selection

Repetition

- *Repeat*



HAPPY CODING

THANK YOU...